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IT Training & Placement

SQL INDUSTRIAL PRACTICAL TASK

Employee Management System – WHERE Clause & Filtering Operators 60 Small Practical Tasks – Basic to Intermediate Mastery

Topics / Activities: Utilize an `employees` table with at least 25 records. Filter by city, department, salary, and joining date. Combine conditions using AND, OR, NOT, BETWEEN, and IN. Practice LIKE patterns and NULL checks.

Faculty Execution: Give requirements first; ask students to identify target column, table, condition, and SQL operator before writing SQL syntax.

Student Tasks & Expected Output: Write and execute filtering queries (Pune employees, salary ranges, IT/HR/Sales lists, name patterns, missing emails, joined conditions). Minimum 15 correctly executed queries required for lab credit.

Suggested target for 1 hour: Minimum Task 1–40, Good Task 1–50, Excellent all 60.

Part A – Table Structure & Initial Verification

Reference structure of populated `employees` table (25+ rows in `company_db`):

Column Name	Data Type & Constraint	Sample Description / Nullability
<code>employee_id</code>	INT PRIMARY KEY	Unique Employee ID (101 to 125+)
<code>employee_name</code>	VARCHAR(100) NOT NULL	Full name of the employee
<code>department</code>	VARCHAR(50)	IT, HR, Sales, Testing, Support, Finance
<code>salary</code>	DECIMAL(10,2)	Monthly compensation (28,000 to 95,000)
<code>city</code>	VARCHAR(50)	Base work location (Pune, Mumbai, Nashik, etc.)
<code>joining_date</code>	DATE	Organization joining date (YYYY-MM-DD)
<code>email</code>	VARCHAR(100) NULL	Corporate email address (contains NULL entries)
<code>status</code>	VARCHAR(20)	Employment state ('Active', 'Inactive')

Part B – Basic Equality & Direct Comparison Filters (=, !=, <>)

- Task 1:** Display all employee records from Pune city.
- Task 2:** Display all employees working in Mumbai.
- Task 3:** Display all employees belonging strictly to the IT department.
- Task 4:** Display all employees belonging to the HR department.
- Task 5:** Display all employees belonging to the Sales department.
- Task 6:** Display all employees whose status is marked as 'Active'.
- Task 7:** Display all employees whose status is marked as 'Inactive'.
- Task 8:** Display the complete record of the employee where `employee_id = 103`.
- Task 9:** Display the record of the employee named 'Priya Sharma'.
- Task 10:** Display all employees whose monthly salary is equal to 35000.
- Task 11:** Display all employees who do not belong to Pune (`city != 'Pune'`).
- Task 12:** Display all employees whose department is NOT Testing using `<>` operator.

Part C – Relational Comparison Operators (>, <, >=, <=)

- Task 13:** Display employees whose salary is greater than 40000.
- Task 14:** Display employees whose salary is less than 35000.
- Task 15:** Display employees whose salary is greater than or equal to 50000.
- Task 16:** Display employees whose salary is less than or equal to 30000.
- Task 17:** Display employees who joined after January 1, 2025 (`joining_date > '2025-01-01'`).
- Task 18:** Display employees who joined on or before December 31, 2024.
- Task 19:** Display employees whose `employee_id` is greater than 110.
- Task 20:** Display employee names and joining dates for employees with `joining_date >= '2026-01-01'` .

Part D – Combining Conditions with Logical AND Operator

- Task 21:** Display Pune employees whose status is Active.
- Task 22:** Display IT department employees whose salary is greater than 50000.
- Task 23:** Display employees located in Mumbai whose status is Inactive.
- Task 24:** Display Sales employees from Pune earning at least 42000.
- Task 25:** Display HR employees who joined after June 1, 2025.
- Task 26:** Display Active employees whose salary is between 40000 and 70000 using `>=` and `<=` with **AND** .
- Task 27:** Display employees from Mumbai in the Testing department with salary `> 38000`.
- Task 28:** Display Active employees who joined on or after '2026-01-01' earning over 45000.

Part E – Logical OR & NOT Operators

- Task 29:** Display employees who belong to either Pune or Mumbai using **OR** .
- Task 30:** Display employees working in the IT department OR HR department.
- Task 31:** Display employees whose salary is strictly less than 32000 OR greater than 60000.
- Task 32:** Display employees who are located in Nashik OR earn more than 55000.
- Task 33:** Display employees who do NOT belong to the HR department using **NOT** (`NOT department = 'HR'`).
- Task 34:** Display employees whose status is NOT Inactive.
- Task 35:** Display employees from Pune OR Mumbai who have Active status (using parentheses).
- Task 36:** Display employees whose city is NOT Pune AND salary is greater than 40000.

Part F – Range & Membership Operators (BETWEEN, IN)

- Task 37:** Display employees whose salary is between 35000 and 55000 using **BETWEEN**.
- Task 38:** Display employees whose salary is NOT between 40000 and 65000 using **NOT BETWEEN**.
- Task 39:** Display employees who joined between '2025-01-01' and '2025-12-31'.
- Task 40:** Display employees whose **employee_id** lies between 105 and 115.
- Task 41:** Display all employees working in 'IT', 'HR', or 'Sales' using the **IN** operator.
- Task 42:** Display employees stationed in 'Pune', 'Mumbai', or 'Nagpur' using **IN**.
- Task 43:** Display employees whose department is NOT IN ('Testing', 'Support').
- Task 44:** Display employees whose city is NOT IN ('Mumbai', 'Nashik').
- Task 45:** Display employees with **employee_id IN (101, 105, 110, 115, 120)**.
- Task 46:** Display IT or Sales employees whose salary is BETWEEN 45000 AND 75000.

Part G – Pattern Matching with LIKE & Wildcards (% , _)

- Task 47:** Display employees whose name starts with 'A' (**LIKE 'A%'**).
- Task 48:** Display employees whose name starts with 'R'.
- Task 49:** Display employees whose name ends with 'a' (**LIKE '%a'**).
- Task 50:** Display employees whose name contains 'sh' anywhere (**LIKE '%sh%'**).
- Task 51:** Display employees whose name starts with 'P' and ends with 'a'.
- Task 52:** Display employees whose name has exactly 5 characters using five underscores (**'_____'**).
- Task 53:** Display employees whose name has 'a' as the second character (**LIKE '_a%'**).
- Task 54:** Display all employees whose name does NOT start with 'R' using **NOT LIKE**.

Part H – NULL Value Checks (IS NULL & IS NOT NULL)

- Task 55:** Display all employees whose email address is missing (**email IS NULL**).
- Task 56:** Display all employees who have a recorded email ID (**email IS NOT NULL**).
- Task 57:** Display employees from Pune whose email address has NOT been provided yet.
- Task 58:** Display Active employees whose email address IS NOT NULL and salary > 40000.

Part I — Combined Industrial Challenge Queries

- Task 59:** Display all Active employees from Pune or Mumbai who belong to the IT or Sales department, earn between 40000 and 70000, and joined after '2025-01-01'.
- Task 60:** Display all employees whose name begins with 'S' or 'R', whose email is NOT NULL, whose status is Active, and whose city is IN ('Pune', 'Nashik').

Topics & Operators Used in This Practical

Relational & Equality Operators

- `=` (Exact Equality Match)
- `!=` or `<>` (Not Equal To)
- `>` and `>=` (Greater Than / Or Equal To)
- `<` and `<=` (Less Than / Or Equal To)
- Date Literal Comparisons (`'YYYY-MM-DD'`)

Logical, Range, Set & Pattern Operators

- `AND` , `OR` , `NOT` (Compound Logic)
- `BETWEEN ... AND ...` (Inclusive Range)
- `IN (...)` & `NOT IN (...)` (Set Membership)
- `LIKE '%...'` , `'_'` (Pattern Matching)
- `IS NULL` & `IS NOT NULL` (Missing Data Checks)

Faculty Checklist & Student Milestone

Milestone	Requirement & Core Concepts	Evaluation Status
Pass Level	Task 1 to 15: Basic Equality, Inequality, and Single Conditions	[] Verified in Lab
Target 1 Hour	Task 1 to 40: AND, OR, NOT combinations and BETWEEN ranges	[] Verified in Lab
Full Practical	Task 1 to 60: Complete set including LIKE wildcards & NULL checks	[] Verified in Lab

Important Query Execution Rules:

- String and Date values in SQL MUST always be enclosed inside single quotes (e.g. `'Pune'` , `'2026-01-10'`). Numerical values must not have quotes.
- In SQL, missing values cannot be compared using equality (`= NULL` is incorrect). Always use `IS NULL` or `IS NOT NULL` .
- When writing complex queries combining `AND` and `OR` , use round parentheses (`...`) to enforce explicit operator precedence.